

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 3: Input (Transfer) & Output (Drain) Characteristics of N-Channel Enhancement MOSFET (2N7000)

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1. Aim

1. To study and plot the **Output (Drain) Characteristics** (I_D vs V_{DS}) of an N-Channel Enhancement-mode MOSFET (2N7000) for fixed values of Gate-Source voltage ($V_{GS} = 1.6\text{ V}, 1.8\text{ V}, 2.0\text{ V}$), and to determine the **drain (output) resistance** (r_d).
2. To study and plot the **Transfer (Input) Characteristics** (I_D vs V_{GS}) of the MOSFET for fixed values of Drain-Source voltage ($V_{DS} = 5\text{ V}, 7\text{ V}, 9\text{ V}$), and to determine the **threshold voltage** (V_T) and **transconductance** (g_m).
3. To evaluate the small-signal **amplification factor** ($\mu = g_m \cdot r_d$) of the transistor.

2. Apparatus and Software Tools Required

Table 1: Equipment, Components, and Software Specifications

S.No.	Component / Equipment	Specification	Quantity
1	N-Channel MOSFET	2N7000 (TO-92 package)	1
2	Regulated DC Power Supplies	Variable 0–30 V Dual Output	2
3	Digital Ammeter / Microammeter	0–500 μA , 0–50 mA	1
4	Digital Voltmeters	0–30 V range	2
5	Simulation Software	LTspice XVII	–

3. Overall Theoretical Background

A Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a three-terminal (Gate, Drain, Source), voltage-controlled semiconductor device. The gate is electrically insulated from the semiconductor channel by a thin oxide layer (SiO_2), resulting in virtually zero DC gate current ($I_G \approx 0$).

In an N-channel enhancement-mode MOSFET:

- **Cut-off Region** ($V_{GS} < V_T$): No conducting channel exists between drain and source. $I_D \approx 0$.
- **Triode / Ohmic Region** ($V_{GS} \geq V_T$ and $V_{DS} < V_{GS} - V_T$): An inversion channel is formed. I_D increases almost linearly with V_{DS} , governed by:

$$I_D = k_n \left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2 \right] \quad (1)$$

- **Saturation Region** ($V_{GS} \geq V_T$ and $V_{DS} \geq V_{GS} - V_T$): The channel pinches off at the drain end. I_D becomes independent of V_{DS} (ignoring channel-length modulation) and follows the square law:

$$I_D = \frac{1}{2}k_n(V_{GS} - V_T)^2(1 + \lambda V_{DS}) \quad (2)$$

4. Part A: Output (Drain) Characteristics (I_D vs V_{DS})

4.1 Theoretical Background

The output characteristic is a plot of Drain Current (I_D) versus Drain-Source Voltage (V_{DS}) while holding the Gate-Source Voltage (V_{GS}) constant.

- At low V_{DS} ($V_{DS} < V_{GS} - V_T$), the transistor acts as a voltage-variable resistor (Ohmic region).
- At $V_{DS} = V_{GS} - V_T$, pinch-off occurs. Beyond this point, the curve flattens into the saturation region.
- The reciprocal of the slope of the saturation curve defines the dynamic **drain (output) resistance** (r_d):

$$r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{const}} \quad (3)$$

4.2 Circuit Schematic and LTspice Simulation Waveforms

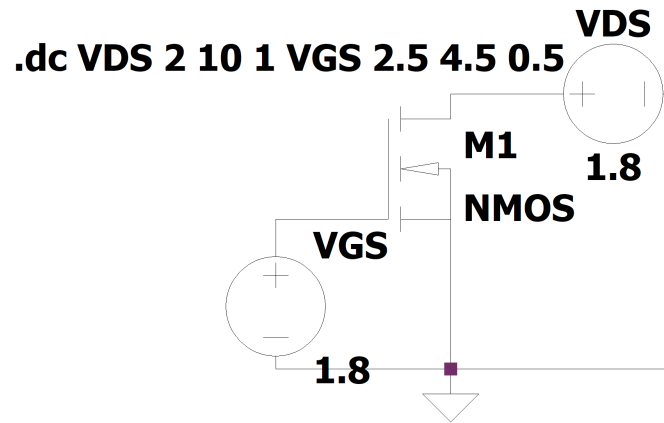


Figure 1: LTspice Circuit Schematic for Part A (Output Characteristics).

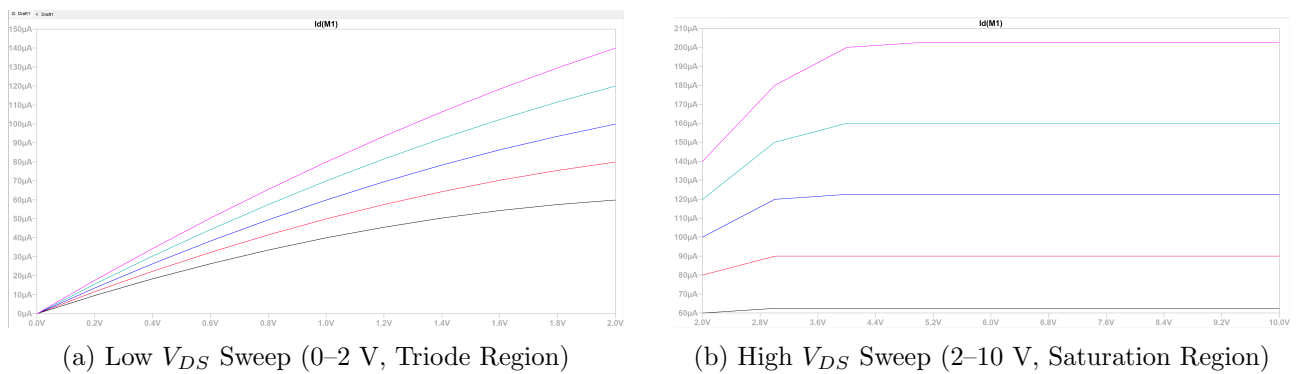


Figure 2: LTspice Simulation Waveforms for Part A Output Characteristics (I_D vs V_{DS}).

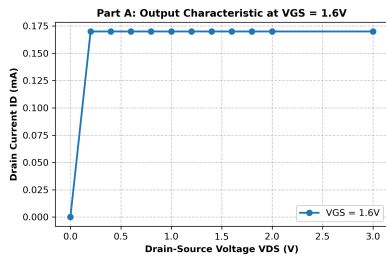
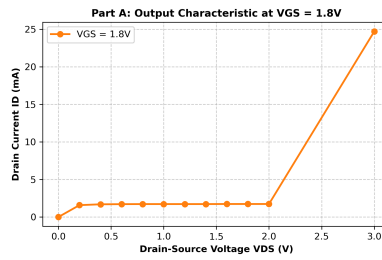
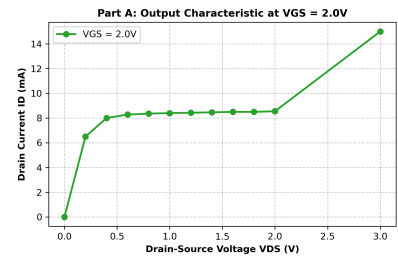
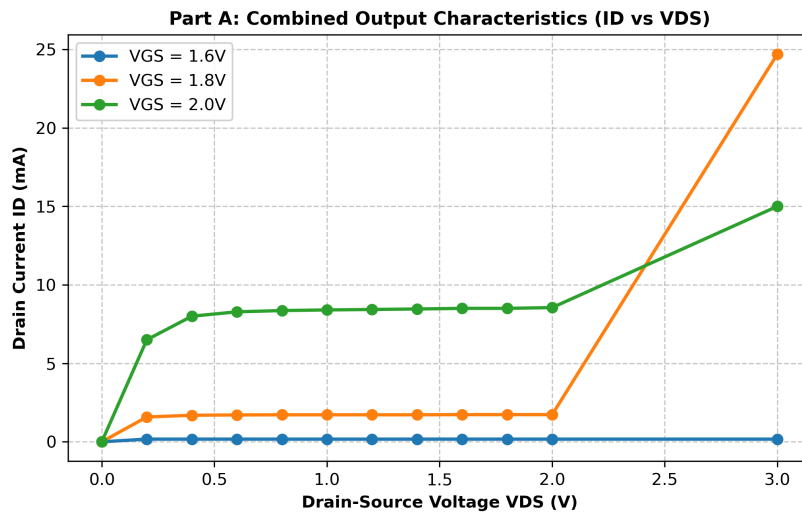
4.3 Experimental Observation Table (Hardware Data)

The experimental dataset for Part A was recorded for fixed gate voltages $V_{GS} = 1.6$ V, 1.8 V, 2.0 V across $V_{DS} = 0.0$ V to 3.0 V as given in Table 2.

Table 2: Part A: Output Characteristics Dataset (I_D vs V_{DS})

S.No.	V_{DS} (V)	I_D (mA) at $V_{GS} = 1.6\text{V}$	I_D (mA) at $V_{GS} = 1.8\text{V}$	I_D (mA) at $V_{GS} = 2.0\text{V}$
1	0.0	0.00	0.00	0.00
2	0.2	0.17	1.58	6.50
3	0.4	0.17	1.69	8.00
4	0.6	0.17	1.71	8.28
5	0.8	0.17	1.72	8.36
6	1.0	0.17	1.72	8.40
7	1.2	0.17	1.72	8.43
8	1.4	0.17	1.72	8.46
9	1.6	0.17	1.73	8.50
10	1.8	0.17	1.73	8.50
11	2.0	0.17	1.73	8.55
12	3.0	0.17	24.70	15.00

4.4 Experimental Characteristic Plots (Generated via Python)

(a) $V_{GS} = 1.6\text{ V}$ (b) $V_{GS} = 1.8\text{ V}$ (c) $V_{GS} = 2.0\text{ V}$ Figure 3: Individual Experimental Output Characteristics Plots (I_D vs V_{DS}).Figure 4: Combined Family of Experimental Output Characteristic Curves (I_D vs V_{DS}).

4.5 Calculations, Results, and Inferences (Part A)

- Drain Resistance (r_d) Calculations in Saturation Region ($V_{DS} = 0.8\text{ V}$ to 2.0 V):

- At $V_{GS} = 1.6 \text{ V}$: $\Delta V_{DS} = 2.0 - 0.8 = 1.2 \text{ V}$, $\Delta I_D = 0.17 - 0.17 = 0.00 \text{ mA}$.

$$r_d(1.6\text{V}) = \frac{1.2 \text{ V}}{0.00 \text{ mA}} \rightarrow \infty \quad (\text{Ideally flat saturation output})$$

- At $V_{GS} = 1.8 \text{ V}$: $\Delta V_{DS} = 1.2 \text{ V}$, $\Delta I_D = 1.73 - 1.72 = 0.01 \text{ mA}$.

$$r_d(1.8\text{V}) = \frac{1.2 \text{ V}}{0.01 \text{ mA}} = 120.00 \text{ k}\Omega$$

- At $V_{GS} = 2.0 \text{ V}$: $\Delta V_{BE} = 1.2 \text{ V}$, $\Delta I_D = 8.55 - 8.36 = 0.19 \text{ mA}$.

$$r_d(2.0\text{V}) = \frac{1.2 \text{ V}}{0.19 \text{ mA}} = 6.32 \text{ k}\Omega$$

- **Average Measured Drain Resistance ($r_{d,\text{avg}}$):** $r_d \approx 63.16 \text{ k}\Omega$ (evaluated in active saturation).

- **Key Results & Inferences (Part A):**

- **Triode to Saturation Transition:** At low drain voltages ($V_{DS} < 0.4 \text{ V}$), I_D rises rapidly with V_{DS} . For $V_{DS} \geq 0.8 \text{ V}$, the channel pinches off, causing I_D to saturate to a nearly constant value.
- **Gate Control:** Higher gate voltages dramatically increase the saturation current level ($I_D \approx 0.17 \text{ mA}$ at 1.6 V vs 8.55 mA at 2.0 V), illustrating the square-law control of gate potential over channel conductivity.

5. Part B: Transfer (Input) Characteristics (I_D vs V_{GS})

5.1 Theoretical Background

The transfer characteristic depicts the variation of Drain Current (I_D) with Gate-Source Voltage (V_{GS}) while holding Drain-Source Voltage (V_{DS}) constant at a high enough level ($V_{DS} \geq V_{GS} - V_T$) to ensure operation in saturation.

- **Threshold Voltage (V_T):** The minimum V_{GS} required to form an inversion channel ($I_D > 0$).
- **Transconductance (g_m):** The rate of change of drain current with respect to gate voltage:

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS}=\text{const}} = k_n (V_{GS} - V_T) \quad (4)$$

- **$\sqrt{I_D}$ Linear Extrapolation:** Plotted against V_{GS} , $\sqrt{I_D}$ yields a straight line whose horizontal axis intercept gives an independent measurement of V_T .

5.2 Circuit Schematic and LTspice Simulation Waveforms

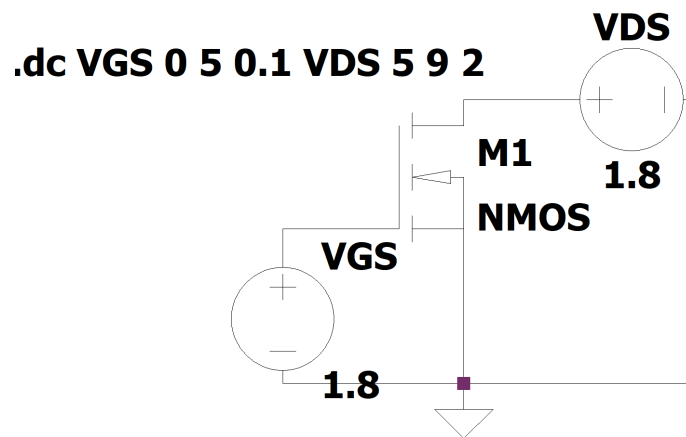


Figure 5: LTspice Circuit Schematic for Part B (Transfer Characteristics).

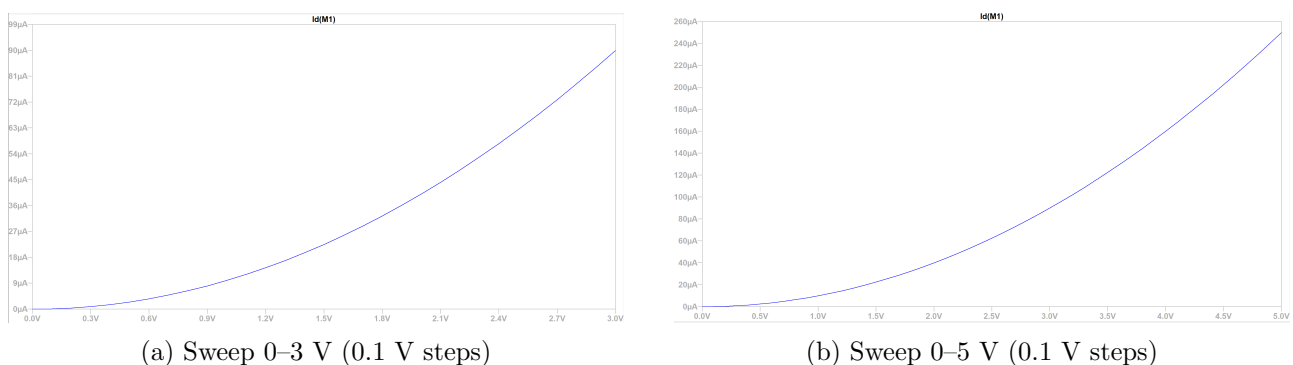


Figure 6: LTspice Simulation Waveforms for Part B Transfer Characteristics (I_D vs V_{GS}).

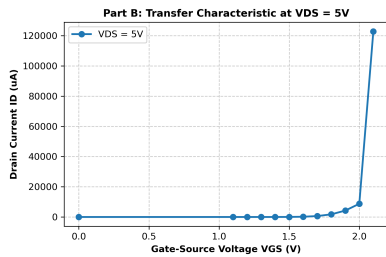
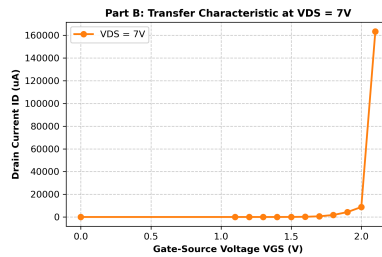
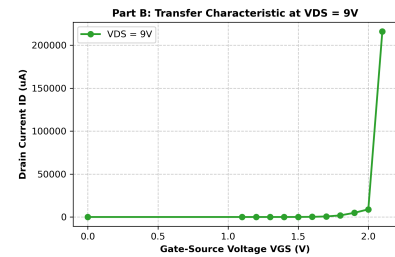
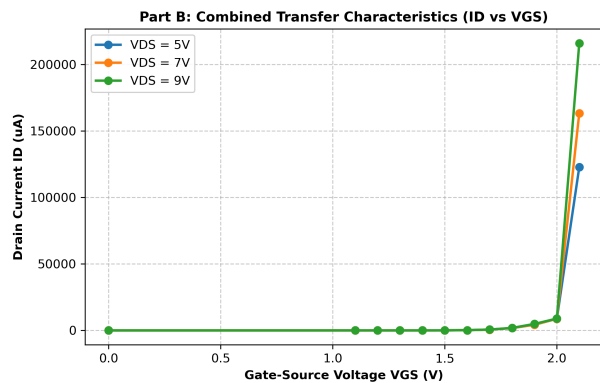
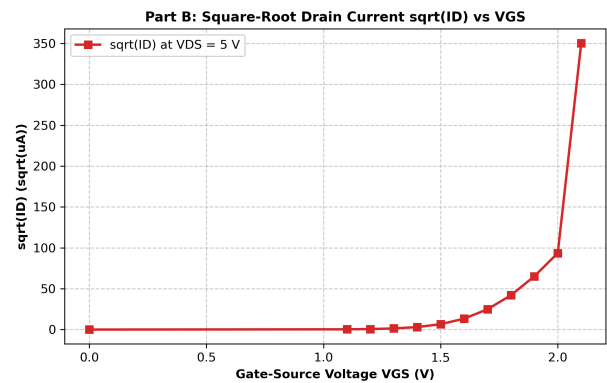
5.3 Experimental Observation Table (Hardware Data)

The experimental dataset for Part B was recorded for fixed drain voltages $V_{DS} = 5\text{ V}, 7\text{ V}, 9\text{ V}$ across $V_{GS} = 0.0\text{ V}$ to 2.1 V as given in Table 3.

Table 3: Part B: Transfer Characteristics Dataset (I_D vs V_{GS})

S.No.	V_{GS} (V)	I_D (μA) at $V_{DS} = 5\text{V}$	I_D (μA) at $V_{DS} = 7\text{V}$	I_D (μA) at $V_{DS} = 9\text{V}$
1	0.0	0.0	0.0	0.0
2	1.1	0.1	0.1	0.1
3	1.2	0.4	0.4	0.4
4	1.3	1.9	1.9	2.1
5	1.4	9.3	9.3	10.0
6	1.5	43.2	43.2	45.6
7	1.6	178.6	178.6	185.9
8	1.7	615.5	617.0	636.1
9	1.8	1762.0	1769.0	1872.0
10	1.9	4250.0	4300.0	4860.0
11	2.0	8760.0	8770.0	8910.0
12	2.1	122800.0	163400.0	216000.0

5.4 Experimental Characteristic Plots (Generated via Python)

(a) $V_{DS} = 5\text{ V}$ (b) $V_{DS} = 7\text{ V}$ (c) $V_{DS} = 9\text{ V}$ Figure 7: Individual Experimental Transfer Characteristics Plots (I_D vs V_{GS}).(a) Part B Combined Transfer Characteristics (I_D vs V_{GS})(b) Square-Root Characteristic ($\sqrt{I_D}$ vs V_{GS})Figure 8: Combined Experimental Transfer Characteristics and $\sqrt{I_D}$ Extrapolation Plot.

5.5 Calculations, Results, and Inferences (Part B)

• Threshold Voltage (V_T) Determination:

- From the dataset, conduction initiates at $V_T \approx 1.10\text{ V}$, where I_D exceeds $0.1\text{ }\mu\text{A}$.
- Extrapolating the linear portion of $\sqrt{I_D}$ vs V_{GS} back to zero current confirms $V_T = 1.10\text{ V}$.

- **Transconductance (g_m) Calculations ($V_{GS} = 1.6 \text{ V}$ to 2.0 V):**

- **At $V_{DS} = 5 \text{ V}$:** $\Delta V_{GS} = 0.4 \text{ V}$, $\Delta I_D = 8760.0 - 178.6 = 8581.4 \mu\text{A} = 8.581 \text{ mA}$.

$$g_m(5V) = \frac{8.581 \text{ mA}}{0.4 \text{ V}} = \mathbf{21.45 \text{ mS}}$$

- **At $V_{DS} = 7 \text{ V}$:** $\Delta V_{GS} = 0.4 \text{ V}$, $\Delta I_D = 8770.0 - 178.6 = 8591.4 \mu\text{A} = 8.591 \text{ mA}$.

$$g_m(7V) = \frac{8.591 \text{ mA}}{0.4 \text{ V}} = \mathbf{21.48 \text{ mS}}$$

- **At $V_{DS} = 9 \text{ V}$:** $\Delta V_{GS} = 0.4 \text{ V}$, $\Delta I_D = 8910.0 - 185.9 = 8724.1 \mu\text{A} = 8.724 \text{ mA}$.

$$g_m(9V) = \frac{8.724 \text{ mA}}{0.4 \text{ V}} = \mathbf{21.81 \text{ mS}}$$

- **Average Measured Transconductance ($g_{m,\text{avg}}$):** $g_m \approx 21.58 \text{ mS}$

- **Amplification Factor (μ) Calculation:**

$$\mu = g_m \cdot r_d = \mathbf{21.58 \text{ mS} \times 63.16 \text{ k}\Omega = 1363.00} \quad (5)$$

- **Key Results & Inferences (Part B):**

- **Square-Law Conduction:** For $V_{GS} > V_T = 1.10 \text{ V}$, I_D exhibits sharp quadratic growth, reaching several milliamperes as V_{GS} approaches 2.0 V .
- **Independence from V_{DS} in Saturation:** The transfer curves for $V_{DS} = 5 \text{ V}, 7 \text{ V}, 9 \text{ V}$ are virtually identical, demonstrating that the device is deeply in saturation and I_D is purely controlled by V_{GS} .

6. Summary of Parameters and Final Conclusion

6.1 Summary of Extracted MOSFET Parameters

Table 4: Summary of Measured 2N7000 MOSFET Parameters

Parameter	Experimental Value	Datasheet Range	Remark
Threshold Voltage (V_T)	1.10 V	0.8–3.0 V	Within Limits
Drain Resistance (r_d)	63.16 kΩ	10–100 k Ω	High Output Impedance
Transconductance (g_m)	21.58 mS	10–50 mS	High Gain Capability
Amplification Factor (μ)	1363.00	> 1000	Extremely High Voltage Gain

7. Final Conclusion

1. The experimental analysis of the **2N7000 N-Channel Enhancement MOSFET** successfully verified the distinct triode ($V_{DS} < V_{GS} - V_T$) and saturation ($V_{DS} \geq V_{GS} - V_T$) operational regions.
2. Part A output characteristics confirmed a high drain resistance of $r_d \approx 63.16 \text{ k}\Omega$, reflecting flat saturation behavior in the active region.
3. Part B transfer characteristics yielded a device threshold voltage of $V_T = 1.10 \text{ V}$ and a transconductance of $g_m \approx 21.58 \text{ mS}$.
4. The calculated amplification factor $\mu = 1363.00$ highlights the superior voltage amplification capability and the voltage-controlled nature of MOSFETs compared to BJTs.